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Differences among Dark Triad Components: A Meta-analytic Investigation

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Abstract

Since its emergence 14 years ago the Dark Triad (DT), composed of narcissism, psychopathy, and Machiavellianism, has become an increasingly popular research focus. Yet questions remain over whether the DT components are sufficiently distinct from another. We examined the nomological networks of each DT component through a meta-analysis of the available literature on the DT. We conducted three separate analyses—an examination of the average intercorrelations among the DT components (*k*=156), an examination of similarities in each DT component's nomological network (*k*=159), and an examination of the effect sizes between DT components and 15 outcome categories (*k* range=7 to 42). Our results indicate that the nomological networks of psychopathy and Machiavellianism overlap substantially while narcissism demonstrated differential relations compared to psychopathy and Machiavellianism. These results remained relatively constant after controlling for DT assessment approach. We argue that the current literature on Machiavellianism may be better understood as a secondary psychopathy literature. Future directions for DT research are discussed in light of our meta-analytic results.

Keywords: Dark Triad, narcissism, psychopathy, Machiavellianism

Differences among Dark Triad Components: A Meta-analytic Investigation

In the last 14 years, a large literature has emerged around three interrelated personality constructs known collectively as the "Dark Triad" (DT). The DT is composed of narcissism, psychopathy, and Machiavellianism all of which have reputations as malevolent personality constructs. Paulhus and Williams (2002) first introduced the three constructs as a distinct cluster of "dark" personalities, and argued that although the DT constructs share a callous, antagonistic core they are distinct from one another. The DT literature has studied the three constructs simultaneously, but each construct has an independent empirical base within psychology.

Narcissism

Narcissism's roots date back thirteen centuries ago to the story of Narcissus, and empirical investigation into the construct has increased in recent decades (Campbell & Miller, 2011; Kenneth, Ellison, & Reynoso, 2012). Narcissism has a robust literature in both social and clinical research domains and is represented by Narcissistic Personality Disorder (NPD) within the American Psychiatric Association's Diagnostic and Statistical Manual (DSM-5; 2013). Research on narcissism has identified two variants: grandiose and vulnerable (Miller et al., 2011; Pincus & Lukowitsky, 2010). Grandiosely narcissistic individuals are characterized by exhibitionism, lack of humility/modesty, and interpersonal dominance, whereas vulnerably narcissistic individuals are characterized by negative affect, distrust, selfishness, and a need for attention and recognition (Dickinson & Pincus, 2003; Miller et al., 2012). Much of the research on narcissism has focused on the grandiose dimension partly due to the fact that the most prominent measure of narcissism—the Narcissistic Personality Inventory (NPI; Raskin & Hall, 1979)—primarily measures features of grandiose narcissism (Miller & Campbell, 2008; Miller, Lynam, & Campbell, 2016). In addition, research on vulnerable narcissism has been conducted mostly within clinical research and not as part of the DT. As a result, narcissism as it is

conceptualized in the DT literature is more reflective of the grandiose variant, although this varies depending on which measure of the DT is used (e.g., Maples, Lamkin, & Miller, 2014).

Psychopathy

Like narcissism, psychopathy has a long history within psychology and has been a part of clinical nomenclature since the mid-19th century (Horley, 2014). The modern conceptualization of the disorder owes much to the theoretical work of Cleckley (1941), Karpman (1941), and the McCords (McCord & McCord, 1964). The impetus for the more recent accumulation of research on psychopathy was Hare's operationalization of the disorder in the Psychopathy Checklist (PCL; 1980) which allowed for empirical measurement of the disorder. Since Hare's initial work, a large number of additional measures have been developed which has contributed to a now substantial research base (see Patrick, 2006 for a review).

Research on psychopathy has been conducted primarily within forensic and clinical psychology due to its robust links with crime and antisocial behavior (Lynam, et al., 2009; Neumann, Hare, & Pardini, 2014). Similar to narcissism, psychopathy is characterized by antagonistic personality traits (e.g. callousness, lack of empathy, and meanness) but differs from narcissism in its relations with disinhibition, which features prominently in nearly all conceptions of psychopathy (e.g. Cleckley, 1941; Hare, 1980). Of the DT constructs, psychopathy is typically considered to be the most nefarious (Paulhus & Williams, 2002) due to its relations with more extreme and frequent antisocial behavior. With regard to its clinical representation, psychopathy is explicitly included for the first time in the latest version of the DSM, as an additional specifier for Antisocial Personality Disorder (APD) in the Section III personality disorder trait model in the DSM-5 (APA, 2013).

Machiavellianism

Machiavellianism derives its name from the 16th century writings of Niccolo Machiavelli who described how to best maintain political power through a variety of tactics. Some researchers have also noted that the writings of the ancient Chinese general Sun Tzu (513/1964) contain similar descriptions of Machiavellian behaviors (Jones & Paulhus, 2011). Theoretically, Machiavellian individuals are thought to be strategic and skilled manipulators and callous pragmatists who strive for success while being able to demonstrate average or above average impulse control (Miller, Hyatt, Maples-Keller, Carter, & Lynam, 2015).

Empirical research on Machiavellianism began with Christie and Geis' (Christie & Geis, 1970) development of the MACH-IV, a 20-item self-report instrument whose items were created by Christie through the selection and adaptation of portions of Machiavelli's writings. Although Christie initially developed the measure to examine traits involved in authoritarianism and other socio-political outcomes, research on Machiavellianism has become popular in both social psychology and industrial/organizational (I/O) psychology. Machiavellianism became a focus in I/O psychology (e.g., Gemmil & Heisler, 1972; Siegel, 1973) because it was thought that some characteristics of the construct might be adaptive in particular workplace contexts and/or implicated in white-collar antisocial behavior (Deshong, Grant, & Mullins-Sweatt, 2015; Jonason, Slomski, & Partyka, 2012). Unlike psychopathy and narcissism, Machiavellianism is almost exclusively studied within the social and I/O literature. Indeed, typical conceptualizations of the construct do not associate Machiavellianism with any type of clinical impairment.

Assessment of the Dark Triad

Since Paulhus and Williams' influential article (2002), three primary approaches to DT measurement have emerged. The most common approach to DT measurement, perhaps because it was used in the original DT study, is the use of three independent instruments, typically the

Narcissistic Personality Inventory (NPI; Raskin & Hall, 1979), the Self-Report Psychopathy Scale (SRP-III; Paulhus, Neumann, & Hare, 2009), and the MACH-IV (Christie & Geiss, 1970). The NPI is a 40-item forced choice self-report measure that requires individuals to select one of two sentences that best describes themselves. The NPI is commonly used in non-clinical settings and has been shown to correlate with measures of grandiose narcissism and narcissistic personality disorder (e.g., Miller, Gaughan, Pryor, Kamen, & Campbell, 2009; Miller, Price, Gentile, Lynam, & Campbell, 2012). The first version of the SRP was produced by Hare (1985) and was designed to be a self-report measure based on the PCL. The SRP is currently in its fourth version, but DT research commonly uses the SRP-III, a 64-item measure that assesses a variety of features related to psychopathy. As previously discussed, the MACH-IV is a 20-item self-report instrument created to assess Machiavellian traits and behaviors as measured by endorsement of items originally derived from Machiavelli's writings.

More recently, two DT-specific self-report measures have emerged that measure all three traits within the same inventory: the Dirty Dozen (DD; Jonason & Webster, 2010) and the Short Dark Triad (SD3; Jones & Paulhus, 2014). The authors of the DD originally started with 22 candidate items based on the NPI, SRP, and MACH-IV that were deemed essential to the respective DT constructs. The authors reduced the 22 items to a final 12 items. Thus, the DD assesses each DT construct with 4-item scales in an effort to increase measurement efficiency.

The SD3 is a 27-item measure with 9-item scales for each of the DT constructs. Similar to the goals of the DD, Jones and Paulhus aimed to reduce the total number of items needed to measure the DT, but also attempted to retain adequate coverage of the essential features of each DT construct. The authors began with an item-pool of 41 items and eventually winnowed these down to 27 while still demonstrating adequate coverage of the DT constructs.

Concerns about Discriminative Validity

Over the past 16 years the DT has quickly become a popular focus of research. However, concerns have been raised regarding the degree to which the DT constructs are distinct from one another. Specifically, a point of contention among researchers is whether Machiavellianism, as it has been measured, is truly distinct from psychopathy. This concern was first highlighted nearly four decades ago (Smith & Griffith, 1978) and has been raised by other researchers since then (Glenn & Sellbom, 2015; Miller et al., 2015; McHoskey, Worzel, & Szyarto, 1998; O'Boyle, et al., 2015). Theoretically, psychopathy and Machiavellianism appear relatively distinct. The theoretically meaningful differences between the two constructs lie in psychopathy's positive relation to disinhibition while Machiavellianism is purportedly related to average or above average levels of impulse control (Jones & Paulhus, 2011; Paulhus, 2014; Paulhus & Williams, 2002). Therefore, at least on a theoretical level, the two constructs demonstrate meaningful differences despite their overlap in other areas. Yet the little empirical work that has been done examining this issue suggests that they may not be distinct constructs.

McHoskey, Worszel, and Szyarto (1998) first suggested that the MACH-IV should be understood as a global measure of subclinical psychopathy. After failing to find substantive differences between the MACH-IV and measures of psychopathy in their relations to theoretically meaningful outcomes, they wrote, "We note that personality and social psychologists have been conducting research on psychopathic attributes in noninstitutionalized populations for more than 25 years and that this research is organized under the rubric of Machiavellianism." (pg. 207). McHoskey et al. also offered a compelling argument for the lack of attention to the problem of construct differentiation between psychopathy and Machiavellianism. They argued that the research bases of psychopathy and Machiavellianism

grew in parallel to one another, with clinical psychologists focusing on psychopathy and social and personality psychologists focusing on Machiavellianism with a lack of cross-discipline communication. Thus, the difference between psychopathy and Machiavellianism was implicitly accepted without empirical evidence supporting such a difference.

In a meta-analysis of the DT's relations with FFM domains, O'Boyle and colleagues (2015) found that psychopathy and Machiavellianism could be characterized by largely the same domain level FFM profile. Psychopathy showed moderate negative relations with Agreeableness and Conscientiousness, was unrelated to Neuroticism, and was weakly positively related to Extraversion and Openness. Machiavellianism showed moderate negative relations with Agreeableness and Conscientiousness, was unrelated to Extraversion and Openness, and showed a small positive relation with Neuroticism. As noted by the authors, "the FFM profiles of Machiavellianism and psychopathy were remarkably similar, with each point estimate existing in the 95% confidence interval of the other construct. This, coupled with the strong positive relation between the two constructs…raises concerns about whether these constructs are two sides of the same coin" (p. 652).

Glenn and Sellbom (2015) also investigated the distinctiveness of the DT constructs in an incarcerated sample of adults. Using the Psychopathic Personality Inventory (PPI; Lilienfeld & Andrews, 1996), the NPI, and the MACH-IV the authors investigated the incremental predictive ability of the DT after accounting for the shared variance between the PPI and the other DT measures. The authors found that generally, the DT did not contribute any incremental variance above and beyond the variance accounted for by the PPI. Furthermore, they were unable to conduct a confirmatory factor analysis (CFA) estimating a latent DT factor because such a large percentage of variance among the measures was accounted for by the PPI psychopathy scores.

Miller and colleagues (2015) conducted 3 separate studies—2 of which examined the distinctiveness of psychopathy and Machiavellianism from an empirical perspective, while the third examined their distinctiveness based on theoretical descriptions. Across the studies, Machiavellianism and psychopathy consistently showed convergent empirical relations despite the divergent theoretical profiles provided by expert ratings. Miller et al. ultimately concluded that, "We agree with McHoskey and colleagues (1998) in suggesting that existing measures of Machiavellianism are actually measuring psychopathy and the literature on Machiavellianism is better framed as an alternative literature on psychopathy." (p. 31).

In light of the doubts that have been raised over the distinctiveness of psychopathy and Machiavellianism as they are currently measured, the purpose of the present meta-analysis is to provide a more comprehensive exploration of the empirical networks associated with the DT constructs with a specific interest in the differences and/or similarities between psychopathy and Machiavellianism. Meta-analytic methods are particularly useful for addressing such questions and the current timing seems appropriate as the DT literature is now large enough to be suitable for meta-analysis but is young enough that a summary of its current state can be particularly useful in directing future research. In addition, meta-analytic analyses produce more reliable estimates of effects between variables compared to single studies (Lipsey & Wilson, 2001). Given the previously discussed results that have shown psychopathy and Machiavellianism to be highly similar to one another, we expect that the two constructs will manifest nearly indistinct empirical networks despite the theoretical differences between them.

Method

Although previous meta-analyses on the DT have included research that focused on DT constructs independent of one another (e.g. O'Boyle, Forsyth, Banks, & Story, 2013), we chose

to focus on the recent literature that has emerged since Paulhus and Williams (2002) first introduced the practice of studying the three constructs simultaneously as an overarching “dark” cluster of personality constructs. In order to focus on the specific trend of simultaneous measurement of the DT constructs, we conducted an online search for research articles on the Dark Triad using PsycINFO and the search terms “Dark Triad” and “Dirty Dozen.” Our search returned 192 articles for potential inclusion. Our criteria for inclusion in the meta-analysis were that the article 1) reported empirical research, 2) reported the relations between outcome measures and all three DT constructs and 3) was written in English. The majority of the studies reported zero-order correlations. In cases where only coefficients from multivariate analyses were reported, we contacted authors for the zero-order correlations. Twenty of the twenty-three authors who were contacted provided the requested information. Given the primary question of our meta-analysis, there were no restrictions placed on what type of outcome the article investigated. Although we also did not place any restrictions on the type of sample used in a study (e.g. clinical, college, online, etc.) there were no available clinical or forensic samples that met criteria for inclusion in our study.

After excluding articles that did not include content relevant to the DT and articles that did not use correlational analyses, a total of 129 articles were found suitable for meta-analysis. The 129 articles provided a total of 159 independent samples. If an article reported relevant relations using several samples, each sample was treated as an independent sample. Sample characteristics, the DT measure used in the study (e.g. Dirty Dozen, SD3, NPI-SRP-MACH-IV), zero-order relations, DT scale reliabilities, and DT intercorrelations were recorded for each study. When this information was not reported in the study, we contacted authors in order to obtain it. An independent coder recorded the same information for 20% of the articles included

in the meta-analysis and these results were compared to the original coding done by the first author. For sample characteristics (i.e. sample size, gender %, etc.) the degree of agreement was 99%. Intraclass correlations were used to calculate agreement for the zero-order relations and intercorrelations. The level of agreement for the zero-order relations was .97 and the level of agreement for the DT intercorrelations was .95. Any disagreement between the coders was investigated to ensure the correct relations were used for the meta-analysis.

In cases where only subscales of a specific DT measure were reported (e.g. primary and secondary psychopathy scales from the SRP and not the total score), the correlations for the subscales were averaged to produce an effect size for the total score. Some samples reported correlations for multiple outcomes that were subsets of the same overall construct (e.g. verbal and physical aggression) and these correlations were averaged for the meta-analysis. For studies that used the HEXACO (Ashton & Lee, 2004) as a measure of personality, the correlations reported for the Honesty-Humility and Agreeableness factors were averaged together and included in the Agreeableness category.

Statistical Analyses

Analyses were conducted using the MIX 2.0 (Bax, 2011) meta-analytic program. We conducted three primary analyses using random-effects models (Lipsey & Wilson, 2001). First, using 156 samples (N=52,988; 3 sample intercorrelations could not be obtained and the samples were excluded), we assessed the average intercorrelations between the DT constructs across the samples. Second, using all 159 samples (N=53,826), we assessed the degree of profile convergence among the DT constructs. We also explored whether the degree of profile convergence differed depending on which DT measure was used. To assess the degree of profile convergence, we used the double-entry intraclass correlation (ICC) which takes into account the

shape, elevation, and scatter of the profiles being compared (Furr, 2010).[1] Each DT construct's correlations with all reported sample outcomes were entered into a dataset in which each row represented the relations among the DT with a single outcome variable from a single study and columns represented the three DT constructs. The relations were double-entered and then the double-entered columns were correlated with one another, with larger correlations indicating higher similarity in the nomological network. Third, we extracted 15 outcome categories from the 129 articles (see Table 1 for the full set of outcome variables) for analyses of effect sizes. There were 96 independent samples (N=34,112) that contributed effect sizes to these categories. A minimum of six independent samples were required for an outcome to be included in the analysis. All correlations were transformed using Fisher's z-transformation before analysis. Homogeneity analyses were also conducted in order to test whether there was significant variability in effect sizes within each outcome category. A table with information on the studies included in our analyses is available in the online supplement.

Unpublished data and dissertations were not solicited for the present study. The decision not to seek unpublished data was motivated by the nature of our primary question—does the published research on the DT support the distinctiveness of the three constructs? We believe using published data allows the best test as publication bias in this case should favor studies that do show differentiation as opposed to similarity.

A final consideration for our meta-analysis was whether to correct for imperfect measurement, or unreliability. Typically, researchers choose to disattenuate correlations as the unreliability of measures will produce downwardly biased effect size estimates (Schmidt, Le, &

[1] Although Furr (2010) highlights the shortcomings of the r_{ICC} when used as an omnibus measure of profile similarity, our examination of the each of the components of profile similarity (i.e. shape, elevation, and scatter) showed no evidence of heterogeneous effects among the components which allows the r_{ICC} to be used as a straightforward measure of global profile similarity.

Ilies, 2003). However, our primary research question was concerned with the size of the DT relations with various outcomes *compared to one another*. As such, the concern of imperfect measurement becomes important for our analyses if the DT constructs show differential levels of reliability compared to one another. However, the comparison of each of the DT reliabilities show that there were no substantive differences in how reliably any of the DT constructs were measured (See Tables 1-4).

Results

Average Intercorrelations

Averaging across measures and across 156 samples, DT constructs were moderately to highly correlated: *r*s = .35 for Machiavellianism and narcissism, .38 for narcissism and psychopathy, and .52 for Machiavellianism and psychopathy. The degree of intercorrelations among the constructs was relatively stable across DT measures; corresponding correlations were .26, .42, and .51 for the single dimension measures (e.g., NPI, SRP, and MACH IV); .49, .33, and .53 for the DD; and .26, .35, and .52 for the SD3. The sole exception to the pattern was the large correlation (r = .49) between DD Machiavellianism and DD narcissism.

Profile Similarity

The second step in our analyses was to examine how similar the DT constructs were in their nomological networks, or the relations they bear to the criterion variables reported across 159 independent samples. The profiles show moderate to high degrees of convergence across all DT measures (narcissism and psychopathy r_{ICC} =.60, narcissism and Machiavellianism r_{ICC} =.57, Machiavellianism and psychopathy r_{ICC} =.86). The particularly high degree of overlap in the empirical networks of Machiavellianism and psychopathy suggests that the two constructs are

nearly indistinguishable in their empirical relations.[2]

We also examined whether profile similarity differed depending on which DT measure was used in a study. The degree of convergence between narcissism and the other DT constructs did appear to vary as a function of DT measure. The NPI and SRP showed the highest degree of profile similarity (r_{ICC} =.66) compared to the narcissism and psychopathy scales of the DD and SD3 (r_{ICCs} =.54 and .55, respectively). The similarity between narcissism and Machiavellianism was also dependent upon DT measure. The profile similarity between the Machiavellianism and narcissism scales from the DD was highest (r_{ICC} =.71) and noticeably larger than the degree of similarity found for the SD3 Machiavellianism and narcissism scales (r_{ICC} =.47) and the NPI and MACH-IV (r_{ICC} =.51). The similarity between psychopathy and Machiavellianism was consistent across measures (r_{ICCs} = .86 for all measurement approaches).

Effect Size Estimates

The final step of our analyses involved sorting the various outcomes from the 159 available samples into fifteen different criterion categories to more specifically examine the similarities and differences among the effect sizes for the DT constructs in relation to theoretically relevant criteria. To this end, we identified 15 broad categories: domains of the Big Five, aggression, impulsivity, sensation seeking/risk taking, emotional intelligence, altruism/empathy, social conservatism, authoritarianism, social dominance, internalizing symptoms, and short-term mating behaviors. The weighted correlations between the DT constructs and the fifteen categories can be seen in in Table 1 with their respective ninety-five percent confidence intervals (CI).

[2] As a comparison, Lynam, Miller, & Derefinko (in press) provided the similarities in the correlational profiles of the NEO PI-R facets for 7 different psychopathy indices. The similarities ranged from .54 to .97 with a median of .86. Thus, Machiavellianism and psychopathy in the present study are as similar to each other as indicators of psychopathy are to one another.

In terms of relations to the Big Five, narcissism was weakly positively correlated with Openness, and weakly to moderately correlated with Extraversion (positively) and Agreeableness (negatively), and unrelated to Neuroticism and Conscientiousness. In contrast, both psychopathy and Machiavellianism were uncorrelated with Extraversion, Neuroticism, and Openness, moderately to strongly correlated negatively with Agreeableness, and weakly to moderately negatively correlated with Conscientiousness. In terms of externalizing behaviors, all three DT constructs were significantly positively correlated with all three outcomes—aggression, impulsivity, and sensation seeking/risk-taking. However, psychopathy and Machiavellianism were equally strongly related to aggression but more strongly related than was narcissism.

With regard to positive interpersonal outcomes, both psychopathy and Machiavellianism were weakly to moderately negatively related to both emotional intelligence and altruism/empathy; in contrast, narcissism was significantly positively related to emotional intelligence and tended to be less negatively related to altruism/empathy. For social attitudes, none of the three constructs were significantly related to social conservatism, whereas all three were moderately positively related to social dominance. Only psychopathy showed a significant, negative relation to authoritarianism, but its relation did not diverge markedly from narcissism and Machiavellianism's relations. Only Machiavellianism and psychopathy showed positive relations with internalizing symptoms; narcissism showed a small negative relation with internalizing symptoms. Finally, all three constructs were moderately positively related to short-term mating behaviors. The average difference between the effect sizes of psychopathy and Machiavellianism across all outcomes was |.06|, with differences ranging from |.01| to |.14|. The average difference between effect sizes for narcissism and psychopathy was |.17| with a range from |.01| to |.42|. Last, the average difference between narcissism and Machiavellianism was

|.14| with differences ranging from |.01| to |.44|. Importantly, there was significant heterogeneity within effect sizes for 14 of the outcomes as indicated by the *Q* statistic (*Q* range: 29.84 to 633.60). All effect size categories indicated significant heterogeneity with the exception of the effect sizes between narcissism and social dominance.

We conducted separate meta-analyses for each approach to assessing the DT: single dimensions (SD)[3], the Dirty Dozen (DD), and the Short Dark Triad (SD3); this allowed us to examine whether results differed as a function of assessment approach. Tables 2-4 display the effect size statistics for each DT construct with columns representing the effects for the three measures of a particular DT construct. Figure 1 in the supplemental on-line materials reports these results graphically. There were some significant reductions in the *Q* statistic after conducting meta-analyses for each assessment approach. Effect sizes with no significant variability within an outcome are bolded in Tables 2-4. Narcissism's relations with the Big Five differ across measures as the NPI had larger negative relations with Agreeableness compared to the DD-N and SD3-N. Both the NPI and the SD3-N showed larger positive relations with Extraversion compared to the DD-N. The SD3 showed a larger positive relation with Extraversion, and its CI does not include the CI of the NPI or the DD-N. All three narcissism measures showed moderate positive relations with the externalizing outcomes of aggression and sensation seeking/risk-taking and the NPI and DD-N showed moderate positive relations with the impulsivity outcome (the SD3-N did not have enough available studies to compute an effect size). In relation to social outcomes, the SD3-N showed a small, significant positive relation with social conservatism while the NPI and SD3-N were unrelated to authoritarianism. The NPI was

[3] The most common measures used in the single dimension approach were the NPI, SRP-III, and MACH-IV. Of the samples that utilized the single dimension approach, 93% used the NPI, 86% used the SRP-III, and 87% used the MACH-IV.

significantly positively correlated with social dominance. Only the NPI had enough available samples to produce an effect size for the internalizing outcome, and the NPI was unrelated to internalizing symptoms. Last, the NPI and DD-N both displayed moderate positive relations with short-term mating behaviors.

The effect sizes for psychopathy showed a high degree of similarity across measures with the exception of externalizing outcomes. Both the SRP and the SD3-P showed moderate to high positive relations with the externalizing outcomes of aggression, impulsivity, and sensation seeking/risk-taking while the DD-P's relations were smaller in magnitude compared to the other measures. All three psychopathy scales produced similar Big Five profiles characterized by moderate negative relations with Agreeableness and Conscientiousness while being unrelated to Openness, Neuroticism, and Extraversion. The three psychopathy measures also demonstrated moderate negative relations with altruism/empathy. The SRP and SD3-P manifested small negative relations with authoritarianism while the SRP showed a moderate positive relation with social dominance. The SRP was unrelated to internalizing symptoms. Finally, the SRP and DD-P both showed moderate positive relations with short-term mating behaviors.

The Big Five profile of Machiavellianism is similar to that of psychopathy; all three Machiavellianism measures showed moderate negative relations with Agreeableness and Conscientiousness. The MACH-IV and SD3-M showed small negative relations with Extraversion while the DD-M scale was uncorrelated with this domain. All three scales were uncorrelated with Neuroticism, and only the SD3-M was weakly correlated with Openness (negatively) while the other two scales were unrelated to this domain. All three Machiavellianism measures showed small to moderate positive relations with externalizing outcomes. The Machiavellianism measures were also negatively related to altruism/empathy

with the MACH-IV and SD3-M showing moderate negative relations with the outcome while the DD-M showed small negative relations. In relation to social outcomes, the SD3-M showed a small positive relationship with conservatism and was unrelated to authoritarianism as was the MACH-IV. The MACH-IV showed moderate positive relations with social dominance as well as with internalizing symptoms. Last, the MACH-IV and DD-M both showed moderate positive relationships with short-term mating behaviors.

We then tested whether the between-construct effect sizes in Table 1 were statistically different from one another. In addition, we tested whether the effect sizes differed across measurement approach within each DT construct; for example, we asked whether the relation between psychopathy and impulsivity depended upon whether psychopathy was measured using the SRP, the SD3, or the DD. Both moderation analyses were completed using the 'metafor' package (Viechtbauer, 2010) in RStudio (Version 0.99.896; RStudio Team, 2015). The results of the comparisons between construct effect sizes showed that narcissism's relations differed significantly from psychopathy and Machiavellianism's relations for 9 out of 15 outcomes (see subscripts, Table 1). The effect sizes of psychopathy and Machiavellianism differed from one another for only 1 of the 15 outcomes (Impulsivity), and the magnitude of differences was relatively small. For the second moderator analysis examining the influence of DT measurement approach on observed effects within constructs, the results showed that of the 45 comparisons (15 outcomes x 3 constructs), 11 effects significantly differed between DT measures. Of these 11 effects, the majority were Big Five outcomes (8 of the total 11). The remaining differences showed that SD3-N's relation with conservatism was significantly larger than other narcissism measures' relations. For psychopathy, DD-P showed significantly smaller relations with both impulsivity and sensation seeking/risk-taking compared to other psychopathy scales.

Importantly, the relations Machiavellianism showed with externalizing outcomes did not differ significantly across DT measurement approaches.

Discussion

The purpose of the present meta-analysis was to explore the nomological networks of each of the DT constructs using extant measures. We sought to test whether available DT research provided evidence that the constructs were distinct from one another. We were particularly interested in the comparison between the empirical profiles associated with psychopathy and Machiavellianism as questions have been raised regarding the distinctiveness of these two DT constructs. Our results indicate that although there is good evidence that narcissism is meaningfully distinct from Machiavellianism and psychopathy in relation to important outcomes, the same is not true of psychopathy and Machiavellianism. The result from a comparison of empirical profiles suggests a very high degree of overlap in the nomological networks of psychopathy and Machiavellianism—as high as the overlap in nomological networks of alternative measures of psychopathy. Results suggest that the Machiavellianism literature might be better understood as literature on alternative measures of psychopathy.[4]

The MACH-IV is the primary measure when researchers have used the single construct approach. Previous work highlighted problems with the MACH-IV. McHoskey and colleagues (1998) argued that the development of the literature on Machiavellianism as measured by the MACH-IV was never integrated with the psychopathy literature despite their genesis around the same time. Instead, the literatures grew in parallel, with social and personality psychologists focusing on Machiavellianism and clinical and forensic psychologists focusing on psychopathy.

[4] The most popular measures of Machiavellianism (the DD-M and MACH-IV) demonstrated the least amount of divergence from psychopathy. The DD and MACH-IV were used in 77% of the 159 samples included in the profile comparisons.

Indeed, in Christie & Geiss's (1970) work, they spend little to no time discussing how Machiavellianism differed from psychopathy and presented no data to demonstrate the divergent validity of the MACH-IV. Thus, the MACH-IV was not subject to the more rigorous validation procedures that many current measures must complete, specifically in regard to its ability to measure a construct substantively different from psychopathy. McHoskey et al. suggested that psychopathy and MACH-IV Machiavellianism "are essentially the same personality construct." (p. 192) and the current meta-analytic results provide additional support for their argument.

The primary issue arises from the content covered by the MACH-IV. Items on the MACH-IV are related to manipulation (e.g. "The best way to handle people is to tell them what they want to hear"), distrust of others (e.g. "Anyone who completely trusts anyone else is asking for trouble"), and callousness (e.g. "Most people forget more easily the death of their parents than the loss of their property"). There is little content that measures high levels of constraint or the ability to delay gratification to achieve one's goals, content that is crucial to the successful separation of Machiavellian individuals from psychopathic individuals.

The DD's approach to DT measurement has been the subject of investigation in previous research, with critics arguing that the measure lacks discriminant validity and produces smaller relations than expected with theoretically important outcomes (Maples, Lamkin, & Miller, 2014; Miller, Few, et al., 2012). In their discussion of the DD, Miller and colleagues (2012) predicted that the DD-P scale would produce attenuated relations with relevant psychopathy outcomes and our data support their assertion. The attenuated relations observed for impulsivity and sensation seeking/risk-taking outcomes are likely due to the limited scope of the measure, as it assesses a multifaceted construct with only four items—none of which assess disinhibitory traits. In their comparison of the DD-P scale to other available measures of psychopathy, Miller and colleagues

(2012) found that the DD-P scale failed to capture variance related to disinhibition and antagonism that other psychopathy measures were able to account for. Given that disinhibition and antagonism are generally considered core features of psychopathy (Lynam & Miller, 2014; Miller & Lynam, 2015), one explanation for the similarity between the DD-M and DD-P profiles may be due to the DD-P scale lacking adequate coverage of disinhibitory traits. Theory suggests that the primary difference between Machiavellianism and psychopathy lies in psychopathy's positive relations with disinhibition (Jones & Paulhus, 2011; Paulhus & Williams, 2002), while Machiavellianism is theoretically related to average or even above average levels of constraint (Miller et al., 2015). Since the DD-P scale does not adequately assess disinhibitory traits, there is little opportunity for the scales to differentiate from one another.

The SD3 assessment approach produced slightly more promising results than the SD and DD approaches, specifically in regard to the sensation seeking/risk-taking outcome. The SD3-M scale's relation with sensation seeking/risk-taking, while still significant and positive (r =.17), was lower than the relation between the SD3-P scale and sensation seeking/risk-taking (r =.51). In regard to other externalizing outcomes, there was less divergence between the SD3-M and SD3-P for aggression (SD3-P: r =.51, SD3-M: r =.36) and the results of our moderator analyses showed that the SD3-M scale's relations with impulsivity did not significantly differ from the relations observed for the other measures of Machiavellianism. Although the SD3-M scale demonstrates improved divergence from psychopathy in important domains compared to the SD and DD approaches, the SD3-M's empirical profile remains in conflict with its theoretical profile (Miller et al., 2015). Theoretical work on Machiavellianism has posited that the construct should be unrelated to impulsivity and disinhibitory traits and should not manifest small to moderate relations with such outcomes (Paulhus, 2014). As it stands, the SD3-M's relations with relevant

outcomes seems to indicate that it is measuring a less severe form of psychopathy.

One may argue that at the zero-order level, the Machiavellianism and psychopathy show a large degree of overlap but the overlap is minimized after partialling their shared variance. Indeed, a common practice in DT research is to utilize multivariate approaches (i.e. partial out the shared variance between DT constructs) to explore the unique relations of the DT components. When applied to Machiavellianism and psychopathy, it can produce outcomes in line with theory such that Machiavellianism is unrelated to impulsivity while psychopathy shows substantive relations with impulsivity (e.g. Jones & Paulhus, 2011). This practice has been recommended for research on the DT (Furnham, Richards, Rangel, & Jones, 2014) without consideration of potential problems. These problems have been highlighted in the past (Lynam, Hoyle, & Newman, 2006), particularly the difficulty in knowing what is left in a variable after partialling out its shared variance. In the case of the DT, where there is substantial overlap in the constructs, these issues become particularly salient. After partialling the shared variance between psychopathy and Machiavellianism, Miller et al. (2015) note that the only certainty is "that it is not the same construct as the construct with which one began." (p. 32).

Narcissism and the Dark Triad

We were primarily concerned with the empirical profiles of Machiavellianism and psychopathy in our meta-analysis, but our results lend support to the divergent validity of narcissism when compared to the other two DT constructs. At the aggregate level, narcissism consistently demonstrated divergence from psychopathy and Machiavellianism. Narcissism showed divergence both in its nomological network as well as in the effect sizes for Extraversion, Conscientiousness, and aggression such that narcissism was related to higher levels of Extraversion and Conscientiousness, and lower levels of aggression compared to psychopathy

and Machiavellianism.

Differences between narcissism and the other two DT constructs were more noticeable when comparing across measurements approaches. Both the SD3-N scale (*r* =.53) and the NPI (*r* =.34) demonstrated larger positive relations with Extraversion compared to the DD-N (*r* =.18). However, both the SD3-N and DD-N scales had small, negative correlations with Agreeableness (SD3-N: *r* =-.13, DD-N: *r* =-.16) compared to the NPI (*r* =-.34). This result is surprising given that the NPI has been criticized in the past for its use in clinical research with critics arguing that it is a measure of adaptive, sub-clinical narcissism (e.g., Pincus & Lukowitsky, 2010). The SD3-N's particularly strong relationship with Extraversion suggests that its scope may be limited to grandiose narcissism as opposed to vulnerable narcissism, which previous work seems to support (Maples et al., 2014). Furthermore, the SD3-N scale's small, negative relationship with Agreeableness suggests that the scale is primarily focused on an agentic and dominant interpersonal style as opposed to a mixture of dominance and antagonism. Both the SD3-N and the DD-N's attenuated negative relations with Agreeableness run counter to how narcissism has been conceptualized at the trait level, where it is generally understood to be a mixture of dominance (high Extraversion) and antagonism (low Agreeableness). The lower effect sizes found for narcissism in relation to Agreeableness are also likely due to the fact that most of the DT literature uses the Big Five Inventory or related indices to measure Agreeableness, which is known to have less content related to honesty-humility than other measures of the five-factor model (e.g., Miller, Gaughan, Maples, & Price, 2011).

The attenuated relations observed for the DD-N scale for Extraversion and Agreeableness are likely due to the scale's emphasis on the desire for attention and recognition from others. The items contain little content related to interpersonal dominance or antagonism. Maples and

colleagues (2014) found that the DD-N scale captures aspects of both grandiose and vulnerable narcissism, but also found similarly attenuated relations due to the scale's limited breadth.

Limitations

There are some limitations of the present study. First, the DT literature is still relatively young and there is a limited range in the type of samples used in research. Of the 159 samples included in our meta-analysis, 86% (k =137) were either college or MTurk samples. The other 14% of samples were composed of individuals drawn from the community, middle school and high school students, and workplace environments. Future work on the DT should broaden the scope of samples used and will benefit from including samples where the prototypical Machiavellian individual may be more likely to be found (e.g. upper management-level personnel in competitive workplace environments). Similarly, the use of clinical samples may allow for more accurate representation of the prototypical psychopath, where features that make it distinct from Machiavellianism (i.e. disinhibition, low neuroticism) may be better represented.

Second, when effect sizes were compared across DT assessment approaches, we were left in some cases with a small number of samples to examine. Excluding the Big Five domains, the range of samples available for outcome categories ranged from 3 to 7, a lower than ideal number for meta-analysis. As a result, the CIs for each DT measure in Tables 2-4 are much wider compared to the aggregated CIs in Table 3. Furthermore, for some outcomes (e.g. social conservatism) there were only enough samples for one measure of the DT to be examined and thus comparisons between measures were unavailable.

Future Directions

Our results suggest that future research on the DT may need to contend with the fact that current measures of Machiavellianism may be better viewed as measures of psychopathy. This is

particularly important given the growing popularity of the DT within psychological research. As it stands, the empirical networks of the two constructs suggest that research on the DT has fallen victim to the jangle fallacy, which occurs when two constructs are labeled as different but in actuality are identical or nearly identical. Echoing the recommendation of Miller et al. (2015), we suggest that resources aimed at the study of the DT would be best directed at developing and validating a new measure of Machiavellianism that produces an empirical profile consistent with theoretical expectations. Such a profile would then be more distinct from psychopathy and warrant study as a separable construct.

Our results suggest that if researchers were to use a currently available measure of the DT, the SD3 produced the best results compared to other extant measures. However, even the Machiavellian scale of the SD3 produced results that run counter to the theoretical work on Machiavellianism. Namely, the SD3-M produced moderate relations with the externalizing outcomes of sensation seeking/risk-taking and aggression and the two studies that have examined the SD3-M's relations with general measures of impulsivity have produced mixed results. Furthermore, the nomological network similarity between the SD3-M and SD3-P scales was as high as the similarities found for the SD and DD assessment approaches.

Because disinhibitory traits serve as the primary distinction between psychopathy and Machiavellianism, we found it surprising how few studies included variables related to disinhibition and other associated outcomes. Specifically, only 14% of the 159 samples included in our meta-analysis utilized such outcomes. This may help explain why research on the DT has continued to accumulate without much concern paid to whether the constructs were distinct from one another; disinhibitory traits are fairly understudied within DT research. Future work on the DT may benefit from more the inclusion of such outcomes as the literature continues to develop.

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Table 1

Meta-analytic Results for Dark Triad Relations

			Narcissism (α=.79)		Psychopathy (α=.77)		Machiavellianism (α=.75)	
	k	*n*	Weighted Avg ES	95% CI	Weighted Avg ES	95% CI	Weighted Avg ES	95% CI
				Big Five				
Extraversion	41	13587	$.32_a$	.25 to .38	$.02_b$	-.03 to .06	$-.04_b$	-.09 to .01
Agreeableness	42	14011	$-.22_a$	-.28 to -.17	$-.45_b$	-.48 to -.41	$-.43_b$	-.46 to -.39
Conscientiousness	40	13283	$-.01_a$	-.05 to .04	$-.26_b$	-.29 to -.23	$-.21_b$	-.24 to -.18
Neuroticism	42	14135	$.00_a$	-.05 to .05	$.02_a$	-.04 to .08	$.06_a$	.01 to .12
Openness	40	13283	$.14_a$	.09 to .19	$-.03_b$	-.06 to .01	$-.05_b$	-.08 to -.02
				Externalizing				
Aggression	15	4683	$.23_a$	.18 to .28	$.44_b$	.42 to .46	$.39_b$	.35 to .44
Impulsivity	10	4123	$.22_a$	.15 to .28	$.35_b$	.26 to .43	$.23_a$	.17 to .29
SS/Risk-taking	12	5103	$.23_a$	.18 to .28	$.38_b$	.28 to .48	$.24_b$	.18 to .29
				Positive Interpersonal Outcomes				
Emotional Intelligence	7	3044	$.20_a$	.13 to .26	$-.22_b$	-.39 to -.03	$-.24_b$	-.37 to -.09
Altruism/Empathy	11	2203	$-.12_a$	-.20 to -.04	$-.34_b$	-.48 to -.19	$-.28_b$	-.41 to -.14
				Social Attitudes				
Social Conservatism	6	4104	$.03_a$	-.05 to .11	$.02_a$	-.08 to .12	$.03_a$	-.05 to .11
Authoritarianism	8	2068	$.01_a$	-.08 to .11	$-.10_a$	-.18 to -.01	$.00_a$	-.12 to .12
Social Dominance	7	1441	$.26_a$	.21 to .31	$.38_a$	.28 to .47	$.28_a$	.17 to .39
				Internalizing				
Internalizing Symptoms	8	3527	$-.08_a$	-.18 to .02	$.10_b$	-.01 to .20	$.17_b$	.07 to .27
				Short-term Mating				
Short-term Mating Behaviors	11	3856	$.27_a$	.20 to .33	$.32_a$	.26 to .39	$.31_a$	.22 to .39

ES=Effect size; CI=Confidence Interval; the average reliability coefficient for measures of each construct is displayed at the top of each construct column; mismatching subscripts denote that the effect sizes between constructs are statistically different from one another based on meta-regression moderator analyses; MIX 2.0 utilizes the DerSimonian and Laird (1986) method as the weighting procedure for effect sizes in random-effects models.

Table 2

Relations between Single Construct Measures

	Narcissism (α=.82)				Psychopathy (α=.83)				Machiavellianism (α=.74)			
	k	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI
Extraversion	14	3936	**.34**	.28 to .40	14	3936	.07	.00 to .15	14	3936	-.11	-.17 to -.05
Agreeableness	14	4180	-.34	-.41 to -.27	14	4180	-.46	-.52 to -.39	14	4180	-.49	-.55 to -.42
Conscientiousness	13	3632	.00	-.07 to .07	13	3632	-.29	-.35 to -.23	13	3632	-.27	-33 to -.20
Neuroticism	15	4484	-.04	-.12 to .04	15	4484	.06	-.04 to .16	15	4484	.12	.00 to .24
Openness	13	3632	.20	.12 to .28	13	3632	-.01	-.10 to .08	13	3632	-.04	-.09 to .02
Aggression	3	961	.13	-.05 to .30	3	961	.53	.41 to .62	3	961	**.49**	.44 to .54
Impulsivity	4	1278	.26	.12 to .39	4	1278	.44	.36 to .51	4	1278	.20	.04 to .35
SS/Risk-taking	3	1458	.23	.09 to .36	3	1458	.45	.32 to .57	3	1458	.24	.09 to .37
Altruism/Empathy	4	903	-.17	-.31 to -.03	4	903	-.35	-.63 to .01	4	903	-.26	-.55 to .07
Social Conservatism	--	--	--	--	--	--	--	--	--	--	--	--
Authoritarianism	3	443	**-.06**	-.20 to .07	3	443	**-.09**	-.20 to .01	3	443	**-.04**	-.18 to .11
Social Dominance	5	1092	.24	.17 to .31	5	1092	.40	.27 to .52	5	1092	**.34**	.28 to .39
Internalizing Symptoms	6	1919	-.08	-.20 to .04	6	1919	.09	-.05 to .23	6	1919	.21	.08 to .34
Short-term Mating Behaviors	7	2246	.27	.16 to .37	7	2246	.37	.27 to .46	7	2246	.37	.27 to .46

ES=Effect size; CI=Confidence Interval; the average reliability coefficient for measures of each construct is displayed at the top of each column; Bolded effect sizes indicate no significant heterogeneity in effect sizes. Emotional Intelligence was not included as all but one sample used the same measure of the Dark Triad so comparisons were unable to be conducted.

Table 3

Relations between Dirty Dozen Scales

	DD-Narcissism (α=.80)				DD-Psychopathy (α=.70)				DD-Machiavellianism (α=.78)			
	k	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI
Extraversion	17	5166	.18	.11 to .24	17	5166	-.06	-.13 to .01	17	5166	.08	.00 to .16
Agreeableness	17	5166	-.16	-.24 to -.07	17	5166	-.41	-.47 to -.36	17	5166	-.36	-.42 to -.31
Conscientiousness	17	5166	-.06	-.12 to .01	18	5330	-.21	-.25 to -.17	18	5330	-.23	-.28 to -.19
Neuroticism	17	5166	.06	.01 to .11	17	5166	.02	-.07 to .12	17	5166	.05	-.02 to .12
Openness	17	5166	.08	.00 to .16	17	5166	-.05	-.09 to .00	17	5166	-.04	-.09 to .01
Aggression	4	1313	**.24**	.19 to .29	4	1313	**.34**	.27 to .40	4	1313	**.36**	.29 to .43
Impulsivity	4	2396	**.23**	.13 to .32	4	2396	**.22**	.18 to .28	4	2396	**.25**	.19 to .30
SS/Risk-taking	5	2574	.23	.14 to .31	5	2574	.22	.13 to .31	5	2574	.28	.20 to .36
Altruism/Empathy	3	685	-.06	-.14 to .01	3	685	**-.26**	-.33 to -.19	3	685	**-.16**	-.23 to -.09
Social Conservatism	--	--	--	--	--	--	--	--	--	--	--	--
Authoritarianism	--	--	--	--	--	--	--	--	--	--	--	--
Social Dominance	--	--	--	--	--	--	--	--	--	--	--	--
Internalizing Symptoms	--	--	--	--	--	--	--	--	--	--	--	--
Short-term Mating Behaviors	4	1610	**.28**	.22 to .33	4	1610	**.24**	.20 to .29	4	1610	**.24**	.20 to .29

ES=Effect size; CI=Confidence Interval; the average reliability coefficient for measures of each construct is displayed at the top of each column; Bolded effect sizes indicate no significant heterogeneity in effect sizes. Emotional Intelligence was not included as all but one sample used the same measure of the Dark Triad so comparisons were unable to be conducted.

Table 4

Relations between Short Dark Triad Scales

	SD3-Narcissism (α=.70)				SD3-Psychopathy (α=.75)				SD3- Machiavellianism (α=.74)			
	k	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI	*k*	*n*	Weighted Avg ES	95% CI
Extraversion	9	4461	.53	.51 to .56	9	4461	.04	-.01 to .09	9	4461	-.12	-.17 to -.06
Agreeableness	10	4641	-.13	-.19 to -.07	9	4461	-.49	-.53 to -.44	9	4461	-.44	-.48 to -.40
Conscientiousness	9	4461	.08	.03 to .14	9	4461	-.25	-.31 to -.20	9	4461	-.14	-.19 to -.08
Neuroticism	9	4461	-.04	-.18 to .10	9	4461	-.03	-.17 to .10	9	4461	.00	-.11 to .10
Openness	9	4461	.18	.12 to .24	9	4461	-.01	-.06 to .05	9	4461	-.06	-.11 to -.01
Aggression	6	1572	**.22**	.16 to .28	6	1572	**.51**	.45 to .57	6	1572	**.36**	.29 to .42
Impulsivity	--	--	--	--	--	--	--	--	--	--	--	--
SS/Risk-taking	3	847	**.26**	.12 to .38	3	847	.51	.32 to .67	3	847	.17	.00 to .33
Altruism/Empathy	4	615	-.10	-.25 to .05	4	615	-.40	-.62 to -.14	4	615	-.39	-.55 to -.19
Social Conservatism	3	2002	**.10**	.05 to .14	3	2002	.06	-.06 to .18	3	2002	**.11**	.06 to .15
Authoritarianism	3	524	.00	-.25 to .25	3	524	-.10	-.38 to .20	3	524	.01	-.37 to .40
Social Dominance	--	--	--	--	--	--	--	--	--	--	--	--
Internalizing Symptoms	--	--	--	--	--	--	--	--	--	--	--	--
Short-term Mating Behaviors	--	--	--	--	--	--	--	--	--	--	--	--

ES=Effect size; CI=Confidence Interval; the average reliability coefficient for measures of each construct is displayed at the top of each column; Bolded effect sizes indicate no significant heterogeneity in effect sizes. Emotional Intelligence was not included as all but one sample used the same measure of the Dark Triad so comparisons were unable to be conducted.

Table 1

Studies Included in Analyses

Table 1 displays the study characteristics of the papers included in the three primary analyses. Papers with coded outcomes in the "Outcome" column indicate that they contributed at least one effect size to one of the 15 outcomes used in the meta-analysis. If the "Outcome" column is left blank for a study, that indicates that the article only contributed to the intercorrelation and profile convergence analyses. In some cases, samples contributed multiple effect sizes (e.g. a sample that reported correlations between the DT and the Big Five contributed an effect size to each one of the Big Five outcomes).

Citation	N	Sample Type	% Male	% White	Age	Outcomes
Adams, H. M., Luevano, V. X., & Jonason, P. K. (2014). Risky business: Willingness to be caught in an extra-pair relationship, relationship experience, and the Dark Triad. *Personality and Individual Differences, 66*, 204–207. doi:10.1016/j.paid.2014.01.008	210	MTurk	43.3	82	33.57	15
Aghababaei, N., Mohammadtabar, S., & Saffarinia, M. (2014). Dirty Dozen vs. the H factor: Comparison of the Dark Triad and Honesty-Humility in prosociality, religiosity, and happiness. *Personality and Individual Differences, 67*, 6–10. doi:10.1016/j.paid.2014.03.026	223	Employees from two private companies	40.4	0	31.24	1-3, 5
Arvan, M. (2013a). Bad news for conservatives? Moral judgments and the dark triad personality traits: A correlational study. *Neuroethics, 6*(2), 307–318. doi:10.1007/s12152-011-9140-6	1154	MTurk	58	Not given	29 (median)	11
Arvan, M. (2013a). Bad news for conservatives? Moral judgments and the dark triad personality traits: A correlational study. *Neuroethics, 6*(2), 307–318. doi:10.1007/s12152-011-9140-6	567	MTurk	53	Not given	28 (Median)	11
Austin, E. J., & O'Donnell, M. M. (2013). Development and preliminary validation of a scale to assess managing the emotions of others. *Personality and Individual Differences, 55*(7), 834–839. doi:10.1016/j.paid.2013.07.005	695	Web survey respondents	22	Not given	24.3	
Austin, E. J., Saklofske, D. H., Smith, M., & Tohver, G. (2014). Associations of the managing the emotions of others (MEOS) scale with personality, the Dark Triad and trait EI. *Personality and Individual Differences, 65*, 8–13. doi:10.1016/j.paid.2014.01.060	369	University students	33	Not given	18.63	1-5, 9

Austin, E. J., Saklofske, D. H., Smith, M., & Tohver, G. (2014). Associations of the managing the emotions of others (MEOS) scale with personality, the Dark Triad and trait EI. *Personality and Individual Differences, 65*, 8–13. doi:10.1016/j.paid.2014.01.060	432	Web survey respondents	24	Not given	22.25	1-5, 9
Baughman, H. M., Dearing, S., Giammarco, E., & Vernon, P. a. (2012). Relationships between bullying behaviours and the Dark Triad: A study with adults. *Personality and Individual Differences, 52*(5), 571–575. doi:10.1016/j.paid.2011.11.020	657	Community	30	Not given	23.1	6
Baughman, H. M., Jonason, P. K., Lyons, M., & Vernon, P. A. (2014). Liar liar pants on fire: Cheater strategies linked to the Dark Triad. *Personality and Individual Differences, 71*, 35–38. doi:10.1016/j.paid.2014.07.019	462	Undergraduate students	28	Not given	19.49	
Baughman, H. M., Jonason, P. K., Veselka, L., & Vernon, P. A. (2014). Four shades of sexual fantasies linked to the Dark Triad. *Personality and Individual Differences, 67.* doi:10.1016/j.paid.2014.01.034	643	Undergraduate students	29.5	Not given	18.56	15
Brewer, G., Hunt, D., James, G., & Abell, L. (2015). Dark Triad traits, infidelity and romantic revenge. *Personality and Individual Differences, 83*, 122–127. doi:10.1016/j.paid.2015.04.007	102	University students and online survey respondents	0	Not given	22.53	15
Brewer, G., Hunt, D., James, G., & Abell, L. (2015). Dark Triad traits, infidelity and romantic revenge. *Personality and Individual Differences, 83*, 122–127. doi:10.1016/j.paid.2015.04.007	108	University students and online recruitment	0	Not given	20.56	6
Buckels, E. E., Jones, D. N., & Paulhus, D. L. (2013). Behavioral confirmation of everyday sadism. *Psychological Science, 24*(11), 2201–9. doi:10.1177/0956797613490749	71	Undergraduate psychology students	27	Not given	20.37	11
Buckels, E. E., Jones, D. N., & Paulhus, D. L. (2013). Behavioral confirmation of everyday sadism. *Psychological Science, 24*(11), 2201–9. doi:10.1177/0956797613490749	71	Undergraduate psychology students	51	26.8	20.52	9
Buckels, E. E., Trapnell, P. D., & Paulhus, D. L. (2014). Trolls just want to have fun. *Personality and Individual Differences, 67*, 97–102. doi:10.1016/j.paid.2014.01.016	395	Undergraduate psychology students & MTurk	47	Not given	28.62	1-5
Buckels, E. E., Trapnell, P. D., & Paulhus, D. L. (2014). Trolls just want to have fun. *Personality and Individual Differences, 67*, 97–102.	414	MTurk	58	Not given	29.2	

doi:10.1016/j.paid.2014.01.016						
Cairncross, M., Veselka, L., Schermer, J. A., & Vernon, P. a. (2013). A Behavioral Genetic Analysis of Alexithymia and the Dark Triad Traits of Personality. *Twin Research and Human Genetics, 16*(03), 690–697. doi:10.1017/thg.2013.19	464	Community adults	11	Not given	39.27	
Campbell, J., Schermer, J. A., Villani, V. C., Nguyen, B., Vickers, L., & Vernon, P. a. (2009). A behavioral genetic study of the dark triad of personality and moral development. *Twin Research and Human Genetics, 12*(2), 132–136. doi:10.1375/twin.12.2.132	472	Community adults	21	Not given	33.9	
Carter, G. L., Campbell, A. C., & Muncer, S. (2014a). The dark triad personality: Attractiveness to women. *Personality and Individual Differences, 56*(1), 57–61. doi:10.1016/j.paid.2013.08.021	128	Undergraduate students	0 (all women sample)	Not given	19.4	1-5
Carter, G. L., Campbell, A. C., & Muncer, S. (2014b). The dark triad: Beyond a "male" mating strategy. *Personality and Individual Differences, 56*(1), 159–164. doi:10.1016/j.paid.2013.09.001	899	Online recruitment	51	Not given	39.5	1-5, 7-8, 15
Carter, G. L., Montanaro, Z., Linney, C., & Campbell, A. C. (2015). Women's sexual competition and the Dark Triad. *Personality and Individual Differences, 74*, 275–279. doi:10.1016/j.paid.2014.10.022	439	University students and online recruitment	0 (all women sample)	Not given	22.85	
Chabrol, H., Melioli, T., Van Leeuwen, N., Rodgers, R., & Goutaudier, N. (2015). The Dark Tetrad: Identifying personality profiles in high-school students. *Personality and Individual Differences, 83*, 97–101. doi:10.1016/j.paid.2015.03.051	613	French high school students	62	Not given	16.85	14
Crysel, L. C., Cook, C. L., Schember, T. O., & Webster, G. D. (2015). Harry Potter and the measures of personality: Extraverted Gryffindors, agreeable Hufflepuffs, clever Ravenclaws, and manipulative Slytherins. *Personality and Individual Differences, 83*, 174–179. doi:10.1016/j.paid.2015.04.016	128	Harry Potter fans/community	10	Not given	21.6	1-5
Crysel, L. C., Crosier, B. S., & Webster, G. D. (2013). The Dark Triad and risk behavior. *Personality and Individual Differences, 54*(1), 35–40. doi:10.1016/j.paid.2012.07.029	1097	Undergraduate students	35.9	Not given	18.53	7-8

Crysel, L. C., Crosier, B. S., & Webster, G. D. (2013). The Dark Triad and risk behavior. *Personality and Individual Differences, 54*(1), 35–40. doi:10.1016/j.paid.2012.07.029	304	MTurk	55	Not given	32.6	7-8
Deshong, H. L., Grant, D. M., & Mullins-Sweatt, S. N. (2015). Comparing models of counterproductive workplace behaviors : The Five-Factor Model and the Dark Triad. *Personality and Individual Differences, 74*, 55–60. doi:10.1016/j.paid.2014.10.001	163	Undergraduate students	29	74.8	20.89	1-5
Djeriouat, H., & Trémolière, B. (2014). The Dark Triad of personality and utilitarian moral judgment: The mediating role of Honesty/Humility and Harm/Care. *Personality and Individual Differences, 67*, 11–16. doi:10.1016/j.paid.2013.12.026	180	MTurk	37	Not given	37.06	4
Douglas, H., Bore, M., & Munro, D. (2012). Distinguishing the Dark Triad: Evidence from the Five-Factor Model and the Hogan Development Survey. *Psychology, 03*(03), 237–242. doi:10.4236/psych.2012.33033	241	Undergraduate students	22	Not given	22.7	1-6, 10
Egan, V., Chan, S., & Shorter, G. W. (2014). The Dark Triad, happiness and subjective well-being. *Personality and Individual Differences, 67*, 17–22. doi:10.1016/j.paid.2014.01.004.	840	Online recruitment	30	Not given	30.1	1-5
Egan, V., Hughes, N., & Palmer, E. J. (2015). Moral disengagement, the dark triad, and unethical consumer attitudes. *Personality and Individual Differences, 76*, 123–128. doi:10.1016/j.paid.2014.11.054	380	Community adults	28	Not given	37.99	1-5
Fox, J., & Rooney, M. C. (2015). The Dark Triad and trait self-objectification as predictors of men's use and self-presentation behaviors on social networking sites. *Personality and Individual Differences, 76*, 161-165.	800	Nationally representative from Qualtrics	100	73.1	29.29	
*Furtner, M. R., Rauthmann, J. F., & Sachse, P. (2011). The Self-Loving Self-Leader: An Examination of the Relationship Between Self-Leadership and the Dark Triad. *Social Behavior and Personality: An International Journal, 39*(3), 369–379. doi:10.2224/sbp.2011.39.3.369	168	Undergraduate students	26.8	Not given	22.49	
Garcia, D., & Sikström, S. (2014). The dark side of Facebook: Semantic representations of status updates predict the Dark Triad of personality. *Personality and Individual Differences, 67*, 69–74. doi:10.1016/j.paid.2013.10.001	304	MTurk	43	Not given	26.4	1, 4

Giammarco, E. A., & Vernon, P. A. (2014). Vengeance and the Dark Triad: The role of empathy and perspective taking in trait forgivingness. *Personality and Individual Differences, 67*, 23–29. doi:10.1016/j.paid.2014.02.010	219	Undergraduate students	21	Not given	20.56	10
Giammarco, E. a., & Vernon, P. a. (2014). Interpersonal Guilt and the Dark Triad. *Personality and Individual Differences, 81*, 96–101. doi:10.1016/j.paid.2014.01.055	219	Undergraduate students	21	Not given	20.56	14
Giammarco, E. a., Atkinson, B., Baughman, H. M., Veselka, L., & Vernon, P. a. (2013). The relation between antisocial personality and the perceived ability to deceive. *Personality and Individual Differences, 54*(2), 246–250. doi:10.1016/j.paid.2012.09.004	1074	Community adults	35	Not given	22.96	
Giammarco, E. a., Atkinson, B., Baughman, H. M., Veselka, L., & Vernon, P. a. (2013). The relation between antisocial personality and the perceived ability to deceive. *Personality and Individual Differences, 54*(2), 246–250. doi:10.1016/j.paid.2012.09.004	1448	Undergraduate students	34	Not given	18.35	1-5
Goodboy, A. K., & Martin, M. M. (2015). The personality profile of a cyberbully: Examining the Dark Triad. *Computers in Human Behavior, 49*, 1–4. doi:10.1016/j.chb.2015.02.052	227	College students	45.8	Not given	20.97	6
Greitemeyer, T. (2015). Everyday sadism predicts violent video game preferences. *Personality and Individual Differences, 75*, 19–23. doi:10.1016/j.paid.2014.10.049	340	College students and university employees	30	Not given	24.2	1-6
Hodson, G., Hogg, S. M., & MacInnis, C. C. (2009). The role of "dark personalities" (narcissism, Machiavellianism, psychopathy), Big Five personality factors, and ideology in explaining prejudice. *Journal of Research in Personality, 43*(4), 686–690. doi:10.1016/j.jrp.2009.02.005	197	Undergraduate students	20.8	Not given	20.06	11-13
Holtzman, N. S. (2011). Facing a psychopath: Detecting the dark triad from emotionally-neutral faces, using prototypes from the Personality Faceaurus. *Journal of Research in Personality, 45*(6), 648–654. doi:10.1016/j.jrp.2011.09.002	88	Undergraduate students	44	66	19.4	11
Holtzman, N. S., & Strube, M. J. (2013). Above and beyond short-term mating, long-term mating is uniquely tied to human personality.	209	Undergraduate students	44	66	19.4	15

Evolutionary Psychology : An International Journal of Evolutionary Approaches to Psychology and Behavior, *11*(5), 1101–1129.						
Holtzman, N. S., & Strube, M. J. (2013). Above and beyond short-term mating, long-term mating is uniquely tied to human personality. *Evolutionary Psychology : An International Journal of Evolutionary Approaches to Psychology and Behavior*, *11*(5), 1101–1129.	151	Peers of sample 154	43	Not given	20.05	15
Holtzman, N. S., & Strube, M. J. (2013). Above and beyond short-term mating, long-term mating is uniquely tied to human personality. *Evolutionary Psychology : An International Journal of Evolutionary Approaches to Psychology and Behavior*, *11*(5), 1101–1129.	228	MTurk	48	0	25.45	15
Holtzman, N. S., & Strube, M. J. (2013). People with dark personalities tend to create a physically attractive veneer. *Social Psychological and Personality Science*, *4*(4), 461-467.	111	Undergraduate students	36	67	19.35	
Holtzman, N. S., & Strube, M. J. (2013). People with dark personalities tend to create a physically attractive veneer. *Social Psychological and Personality Science*, *4*(4), 461-467.	331	Peers of sample 30	Not given	Not given	19.99	
Jakobwitz, S., & Egan, V. (2006). The dark triad and normal personality traits. *Personality and Individual Differences*, *40*(2), 331–339. doi:10.1016/j.paid.2005.07.006	82	Community adults	37	Not given	29	1-5
James, S., Kavanagh, P. S., Jonason, P. K., Chonody, J. M., & Scrutton, H. E. (2014b). The Dark Triad, schadenfreude, and sensational interests: Dark personalities, dark emotions, and dark behaviors. *Personality and Individual Differences*, *68*, 211–216. doi:10.1016/j.paid.2014.04.020	186	University students and their peers	37.1		31.2	8
Jonason, P. K. (2014). Personality and politics. *Personality and Individual Differences*, *71*, 181–184. doi:10.1016/j.paid.2014.08.002	1389	College students	33	46	18.88	11
Jonason, P. K. (2014). Personality and politics. *Personality and Individual Differences*, *71*, 181–184. doi:10.1016/j.paid.2014.08.002	281	MTurk	36	80	32.26	11
Jonason, P. K. (2015). How “‘ dark ’” personality traits and perceptions come together to predict racism in Australia. *Personality and*	201	University students	17	59	23.24	12-13

Individual Differences, 72, 47–51. doi:10.1016/j.paid.2014.08.030						
Jonason, P. K., & Buss, D. M. (2012). Avoiding entangling commitments: Tactics for implementing a short-term mating strategy. *Personality and Individual Differences, 52*(5), 606–610. doi:10.1016/j.paid.2011.12.015	298	College students	47	Not given	22.84	6, 15
Jonason, P. K., & Kavanagh, P. (2010). The dark side of love: Love styles and the Dark Triad. *Personality and Individual Differences, 49*(6), 606–610. doi:10.1016/j.paid.2010.05.030	302	Online recruitment	19	Not given	28.53	
Jonason, P. K., & Krause, L. (2013). The emotional deficits associated with the Dark Triad traits: Cognitive empathy, affective empathy, and alexithymia. *Personality and Individual Differences, 55*(5), 532–537. doi:10.1016/j.paid.2013.04.027	322	Online sample	25	Not given	24.24	10
Jonason, P. K., & Luévano, V. X. (2013). Walking the thin line between efficiency and accuracy: Validity and structural properties of the Dirty Dozen. *Personality and Individual Differences*, *55*(1), 76-81.	210	MTurk	43	82	33.57	13
Jonason, P. K., & McCain, J. (2012). Using the HEXACO model to test the validity of the Dirty Dozen measure of the Dark Triad. *Personality and Individual Differences, 53*(7), 935–938. doi:10.1016/j.paid.2012.07.010	544	Convenience	31	64	20.25	1-5, 13-14
*Jonason, P. K., & Schmitt, D. P. (2012). What have you done for me lately? Friendship-selection in the shadow of the dark triad traits. *Evolutionary Psychology, 10*(3), 400–421.	114	Online recruitment	24	Not given	24.88	10
Jonason, P. K., & Schmitt, D. P. (2012). What have you done for me lately? Friendship-selection in the shadow of the dark triad traits. *Evolutionary Psychology, 10*(3), 400–421.	267	College students	36	Not given	24.03	10, 15
Jonason, P. K., & Tost, J. (2010). I just cannot control myself: The Dark Triad and self-control. *Personality and Individual Differences, 49*(6), 611–615. doi:10.1016/j.paid.2010.05.031	259	College students	28	Not given	21.12	

Jonason, P. K., & Tost, J. (2010). I just cannot control myself: The Dark Triad and self-control. *Personality and Individual Differences, 49*(6), 611–615. doi:10.1016/j.paid.2010.05.031	96	Undergraduate students	27	Not given	19.53	
Jonason, P. K., & Webster, G. D. (2010). The dirty dozen: a concise measure of the dark triad. *Psychological Assessment, 22*(2), 420–432. doi:10.1037/a0019265	273	Undergraduate students	32	Not given	20.08	1-5, 15
Jonason, P. K., & Webster, G. D. (2010). The dirty dozen: a concise measure of the dark triad. *Psychological Assessment, 22*(2), 420–432. doi:10.1037/a0019265	246	Undergraduate students	41	Not given	20.69	1-5, 15
Jonason, P. K., & Webster, G. D. (2010). The dirty dozen: a concise measure of the dark triad. *Psychological Assessment, 22*(2), 420–432. doi:10.1037/a0019265	60	Undergraduate students	38	Not given	20.44	6
Jonason, P. K., & Webster, G. D. (2012). A protean approach to social influence: Dark Triad personalities and social influence tactics. *Personality and Individual Differences, 52*(4), 521–526. doi:10.1016/j.paid.2011.11.023	259	Undergraduate students	28	Not given	21.12	
Jonason, P. K., & Webster, G. D. (2012). A protean approach to social influence: Dark Triad personalities and social influence tactics. *Personality and Individual Differences, 52*(4), 521–526. doi:10.1016/j.paid.2011.11.023	296	Undergraduate students	45	Not given	23.02	
Jonason, P. K., & Webster, G. D. (2012). A protean approach to social influence: Dark Triad personalities and social influence tactics. *Personality and Individual Differences, 52*(4), 521–526. doi:10.1016/j.paid.2011.11.023	268	Undergraduate students	45	Not given	21.19	
Jonason, P. K., Baughman, H. M., Carter, G. L., & Parker, P. (2015). Dorian Gray without his portrait: Psychological, social, and physical health costs associated with the Dark Triad. *Personality and Individual Differences, 78*, 5–13. doi:10.1016/j.paid.2015.01.008	1389	American undergraduate students	33	46	18.88	14
Jonason, P. K., Baughman, H. M., Carter, G. L., & Parker, P. (2015). Dorian Gray without his portrait: Psychological, social, and physical health costs associated with the Dark Triad. *Personality and Individual Differences, 78*, 5–13. doi:10.1016/j.paid.2015.01.008	280	British undergraduate students	16	Not given	Not given	8

Jonason, P. K., Duineveld, J. J., & Middleton, J. P. (2015). Pathology, pseudopathology, and the Dark Triad of personality. *Personality and Individual Differences, 78*, 43–47. doi:10.1016/j.paid.2015.01.028	290	Recruited from social networking sites	35	Not given	31.16	6
Jonason, P. K., Jones, A., & Lyons, M. (2013). Creatures of the night: Chronotypes and the Dark Triad traits. *Personality and Individual Differences, 55*(5), 538–541. doi:10.1016/j.paid.2013.05.001	263	Online recruitment	28	Not given	24.72	
Jonason, P. K., Koenig, B. L., & Tost, J. (2010). Living a Fast Life: The Dark Triad and Life History Theory. *Human Nature, 21*(4), 428–442. doi:10.1007/s12110-010-9102-4	246	Undergraduate students	41	Not given	20.69	
Jonason, P. K., Koenig, B. L., & Tost, J. (2010). Living a Fast Life: The Dark Triad and Life History Theory. *Human Nature, 21*(4), 428–442. doi:10.1007/s12110-010-9102-4	321	Undergraduate students	31	Not given	20.81	15
Jonason, P. K., Li, N. P., & Buss, D. M. (2010). The costs and benefits of the Dark Triad: Implications for mate poaching and mate retention tactics. *Personality and Individual Differences, 48*(4), 373–378. doi:10.1016/j.paid.2009.11.003	336	Online recruitment	34	Not given	27	15
Jonason, P. K., Li, N. P., & Teicher, E. a. (2010). Who is James Bond?: The dark triad as an agentic social style. *Individual Differences Research, 8*(2), 111–120.	216	Upper division and weekend psychology students	37	Not given	22.51	1-5
Jonason, P. K., Li, N. P., & Teicher, E. a. (2010). Who is James Bond?: The dark triad as an agentic social style. *Individual Differences Research, 8*(2), 111–120.	236	Online recruitment	34	Not given	26.65	10
Jonason, P. K., Li, N. P., & Webster., G. D. (2009). The Dark Triad: Facilitating a Short-Term Mating in Men. *European Journal of Personality, 23*(1), 5–18.	224	Undergraduate students	39	Not given	23.5	15
Jonason, P. K., Li, N. P., & Czarna, A. Z. (2013). Quick and dirty: Some psychosocial costs associated with the dark triad in three countries. *Evolutionary Psychology, 11*(1), 172–185.	325	American undergraduate students	58	Not given	21.17	10
Jonason, P. K., Li, N. P., & Czarna, A. Z. (2013). Quick and dirty: Some psychosocial costs associated with the dark triad in three countries. *Evolutionary Psychology, 11*(1), 172–	188	Singaporean undergraduate students	41	Not given	21.89	10

185.						
Jonason, P. K., Li, P. C. N., & Czarna, A. (2013). Quick and dirty: Some psychosocial costs associated with the Dark Triad in three countries. *Evolutionary Psychology, 11*(1), 172-185.	175	Polish undergraduate students	38	Not given	26.99	10
Jonason, P. K., Luevano, V. X., & Adams, H. M. (2012). How the Dark Triad traits predict relationship choices. *Personality and Individual Differences, 53*(3), 180–184. doi:10.1016/j.paid.2012.03.007	210	MTurk	43	83.8	33.57	15
Jonason, P. K., Lyons, M., & Bethell, E. (2014). The making of Darth Vader: Parent-child care and the Dark Triad. *Personality and Individual Differences, 67*(July 2015), 30–34. doi:10.1016/j.paid.2013.10.006	352	University students and community members	17	Not given	25.1	
Jonason, P. K., Lyons, M., Baughman, H. M., & Vernon, P. a. (2014). What a tangled web we weave: The Dark Triad traits and deception. *Personality and Individual Differences, 70*, 117–119. doi:10.1016/j.paid.2014.06.038	447	University students	36	Not given	23.47	
Jonason, P. K., Slomski, S., & Partyka, J. (2012). The Dark Triad at work: How toxic employees get their way. *Personality and Individual Differences, 52*(3), 449–453. doi:10.1016/j.paid.2011.11.008	419	Online recruitment	30	Not given	22.78	13
Jonason, P. K., Strosser, G. L., Kroll, C. H., Duineveld, J. J., & Baruffi, S. a. (2015). Valuing myself over others: The Dark Triad traits and moral and social values. *Personality and Individual Differences, 81*, 102–106. doi:10.1016/j.paid.2014.10.045	585	MTurk	46	78	33.36	13
Jonason, P. K., Strosser, G. L., Kroll, C. H., Duineveld, J. J., & Baruffi, S. a. (2015). Valuing myself over others: The Dark Triad traits and moral and social values. *Personality and Individual Differences, 81*, 102–106. doi:10.1016/j.paid.2014.10.045	252	American college students	38	87	20.67	13
Jonason, P. K., Strosser, G. L., Kroll, C. H., Duineveld, J. J., & Baruffi, S. a. (2015). Valuing myself over others: The Dark Triad traits and moral and social values. *Personality and Individual Differences, 81*, 102–106. doi:10.1016/j.paid.2014.10.045	516	German college students	35	Not given	23.99	5, 13

Jonason, P. K., Valentine, K. a., Li, N. P., & Harbeson, C. L. (2011). Mate-selection and the Dark Triad: Facilitating a short-term mating strategy and creating a volatile environment. *Personality and Individual Differences, 51*(6), 759–763. doi:10.1016/j.paid.2011.06.025	242	Undergraduate students	44	Not given	20.89	10
Jonason, P. K., Wee, S., & Li, N. P. (2015). Competition, autonomy, and prestige: Mechanisms through which the Dark Triad predict job satisfaction. *Personality and Individual Differences, 72*, 112–116. doi:10.1016/j.paid.2014.08.026	361	MTurk	56	Not given	33.7	
Jonason, P. K., Kaufman, S. B., Webster, G. D., & Geher, G. (2013). What lies beneath the Dark Triad Dirty Dozen: Varied relations with the Big Five. *Individual Differences Research, 11*(2), 81–90.	123	Undergraduate students	24	Not given	20.35	1-5
Jonason, P. K., Kaufman, S. B., Webster, G. D., & Geher, G. (2013). What lies beneath the Dark Triad Dirty Dozen: Varied relations with the Big Five. *Individual Differences Research, 11*(2), 81–90.	290	Undergraduate students	41	52	22.42	1-5, 10
Jones, D. N. (2013a). Psychopathy and machiavellianism predict differences in racially motivated attitudes and their affiliations. *Journal of Applied Social Psychology, 43*(SUPPL.2), 367–378. doi:10.1111/jasp.12035	157	MTurk	38	82	33.5	11-12
Jones, D. N. (2013a). Psychopathy and machiavellianism predict differences in racially motivated attitudes and their affiliations. *Journal of Applied Social Psychology, 43*(SUPPL.2), 367–378. doi:10.1111/jasp.12035	89	MTurk	36	100	35.2	11-12
Jones, D. N. (2013b). What's mine is mine and what's yours is mine: The Dark Triad and gambling with your neighbor's money. *Journal of Research in Personality, 47*(5), 563–571. doi:10.1016/j.jrp.2013.04.005	119	MTurk	46	67	31.91	
Jones, D. N. (2013b). What's mine is mine and what's yours is mine: The Dark Triad and gambling with your neighbor's money. *Journal of Research in Personality, 47*(5), 563–571. doi:10.1016/j.jrp.2013.04.005	109	MTurk	52	76	33.21	
Jones, D. N. (2014). Risk in the face of retribution: Psychopathic individuals persist in financial misbehavior among the Dark Triad. *Personality and Individual Differences, 67,*	237	MTurk	50.2	69	30.88	

109–113. doi:10.1016/j.paid.2014.01.030						
Jones, D. N., & Figueredo, A. J. (2013). The Core of Darkness: Uncovering the Heart of the Dark Triad. *European Journal of Personality, 27*(6), 521–531. doi:10.1002/per.1893	388	MTurk	46	56	31.28	13
Jones, D. N., & Olderbak, S. G. (2014). The associations among dark personalities and sexual tactics across different scenarios. *Journal of interpersonal violence, 29*(6), 1050-1070.	186	MTurk	100	67	30.56	13
Jones, D. N., & Olderbak, S. G. (2013). The associations among dark personalities and sexual tactics across different scenarios. *Journal of Interpersonal Violence, 29*(6), 1050–70. doi:10.1177/0886260513506053	261	MTurk	100	72	29.95	13
Jones, D. N., & Paulhus, D. L. (2011). The role of impulsivity in the Dark Triad of personality. *Personality and Individual Differences, 51*(5), 679–682. doi:10.1016/j.paid.2011.04.011	142	Undergraduate students	28	43	20.35	7
Jones, D. N., & Paulhus, D. L. (2011). The role of impulsivity in the Dark Triad of personality. *Personality and Individual Differences, 51*(5), 679–682. doi:10.1016/j.paid.2011.04.011	329	MTurk	38	60	29.68	7
Jones, D. N., & Paulhus, D. L. (2014). Introducing the short Dark Triad (SD3): a brief measure of dark personality traits. *Assessment, 21*(1), 28–41. doi:10.1177/1073191113514105	230	MTurk	42	62	28.9	13
Jones, D. N., & Weiser, D. A. (2014). Differential infidelity patterns among the Dark Triad. *Personality and Individual Differences, 57*, 20–24. doi:10.1016/j.paid.2013.09.007	884	MTurk	51	60	30.53	15
Kajonius, P. J., Persson, B. N., & Jonason, P. K. (2015). Hedonism, Achievement, and Power: Universal values that characterize the Dark Triad. *Personality and Individual Differences, 77*, 173-178. doi:10.1016/j.paid.2014.12.055	385	Undergraduate students/MTurk	39	Not given	31.85	10
Kämmerle, M., Unterrainer, H. F., Dahmen-Wassenberg, P., Fink, A., & Kapfhammer, H. P. (2014). Dimensions of Religious/Spiritual Well-Being and the Dark Triad of Personality. *Psychopathology, 47*(5), 297-302. doi:10.1159/000358563	312	German university students	29	Not given	23.76	

Kardum, I., Hudek-Knezevic, J., Schmitt, D. P., & Grundler, P. (2015). Personality and mate poaching experiences. *Personality and Individual Differences, 75*, 7-12. doi:10.1016/j.paid.2014.10.048	819	University students	49	Not given	20.96	1-5, 15
Kavanagh, P. S., Signal, T. D., & Taylor, N. (2013). The Dark Triad and animal cruelty: Dark personalities, dark attitudes, and dark behaviors. *Personality and Individual Differences, 55*(6), 666-670. doi:10.1016/j.paid.2013.05.019	227	Online recruitment	9	Not given	30.1	
Klimstra, T.A., Sijtsema, J.J., Henrichs, J., & Cima, M. (2014). The Dark Triad of personality in adolescence: Psychometric properties of concise measure and associations with adolescent adjustment from a multi-informant perspective. *Journal of Research in Personality, 53*, 84-92. doi:10.1016/j.jrp.2014.09.001	611	High-school adolescents	47.8		16.79	1-6
Klimstra, T.A., Sijtsema, J.J., Henrichs, J., & Cima, M. (2014). The Dark Triad of personality in adolescence: Psychometric properties of concise measure and associations with adolescent adjustment from a multi-informant perspective. *Journal of Research in Personality, 53*, 84-92. doi:10.1016/j.jrp.2014.09.001	302	Adolescents	51.7		12.79	1-6
Lee, K., & Ashton, M. C. (2005). Psychopathy, Machiavellianism, and narcissism in the Five-Factor Model and the HEXACO model of personality structure. *Personality and Individual Differences, 38*(7), 1571-1582. doi:10.1016/j.paid.2004.09.016	164	Undergraduate students	44	Not given	23.9	1-5
Lee, K., Ashton, M. C., Wiltshire, J., Bourdage, J. S., Visser, B. A., & Gallucci, A. (2013). Sex, power, and money: Prediction from the Dark Triad and Honesty–Humility. *European Journal of Personality, 27*(2), 169-184. doi:10.1002/per	232	Undergraduate students	29	Not given	21	1-5
Lee, K., Ashton, M. C., Wiltshire, J., Bourdage, J. S., Visser, B. A., & Gallucci, A. (2013). Sex, power, and money: Prediction from the Dark Triad and Honesty–Humility. *European Journal of Personality, 27*(2), 169-184. doi:10.1002/per	232	Undergraduate students	Not given	Not given	Not given	1-5
Lee, K., Ashton, M. C., Wiltshire, J., Bourdage, J. S., Visser, B. A., & Gallucci, A. (2013). Sex, power, and money: Prediction from the Dark Triad and Honesty–Humility. *European Journal of Personality, 27*(2), 169-184. doi:10.1002/per	200	Undergraduate students	33	Not given	21	1-5

Lee, K., Ashton, M. C., Wiltshire, J., Bourdage, J. S., Visser, B. A., & Gallucci, A. (2013). Sex, power, and money: Prediction from the Dark Triad and Honesty–Humility. *European Journal of Personality*, *27*(2), 169-184. doi:10.1002/per	200	Undergraduate students	Not given	Not given	Not given	1-5
Linton, D. K., & Power, J. L. (2013). The personality traits of workplace bullies are often shared by their victims: Is there a dark side to victims?. *Personality and Individual differences*, *54*(6), 738-743. doi:10.1016/j.paid.2012.11.026	224	Undergraduate students	29.5	Not given	21.45	6-8
Lodi-Smith, J., Shepard, K., & Wagner, S. (2014). Personality and sexually deviant behavior. *Personality and Individual Differences*, *70*, 39-44. doi:10.1016/j.paid.2014.06.012	595	Online recruitment	41	87.9	23.64	15
Lyons, M. T., & Hughes, S. (2015). Malicious mouths? The Dark Triad and motivations for gossip. *Personality and Individual Differences*, *78*, 1-4. doi:10.1016/j.paid.2015.01.009	372	University students/community members	20	Not given	24.88	
Lyons, M., & Rice, H. (2014). Thieves of time? Procrastination and the Dark Triad of personality. *Personality and Individual Differences*, *61*, 34-37. doi:10.1016/j.paid.2014.01.002	369	Online recruitment	31	Not given	28.34	
Maples, J. L., Lamkin, J., & Miller, J. D. (2014). A test of two brief measures of the dark triad: The dirty dozen and short dark triad. *Psychological assessment*, *26*(1), 326-331. doi:10.1037/a0035084	287	MTurk	51.2	80.5	34.3	1-5
*Marcus, D. K., Zeigler-Hill, V., Mercer, S. H., & Norris, A. L. (2014). The psychology of spite and the measurement of spitefulness. *Psychological assessment*, *26*(2), 563-574. doi:10.1037/a0036039	556	Undergraduate students	21	74.4	19.9	
*Marcus, D. K., Zeigler-Hill, V., Mercer, S. H., & Norris, A. L. (2014). The psychology of spite and the measurement of spitefulness. *Psychological assessment*, *26*(2), 563-574. doi:10.1037/a0036039	297	MTurk	38	75.4	36.4	
Martin, R. A., Lastuk, J. M., Jeffery, J., Vernon, P. A., & Veselka, L. (2012). Relationships between the Dark Triad and humor styles: A replication and extension. *Personality and Individual Differences*, *52*(2), 178-182. doi:10.1016/j.paid.2011.10.010	200	Undergraduate students	46	Not given	20.7	6, 10

McCain, J. L., Jonason, P. K., Foster, J. D., & Campbell, W. K. (2015). The bifactor structure and the "dark nomological network" of the State Self-Esteem Scale. *Personality and Individual Differences, 72*, 1-6. doi:10.1016/j.paid.2014.08.006	544	University students	31	64	20.25	
McDonald, M. M., Donnellan, M. B., & Navarrete, C. D. (2012). A life history approach to understanding the Dark Triad. *Personality and Individual Differences, 52*(5), 601-605. doi:10.1016/j.paid.2011.12.003	296	College students	51	84.1	Not given	6
McLarnon, M. J., Carswell, J. J., & Schneider, T. J. (2015). A case of mistaken identity? Latent profiles in vocational interests. *Journal of Career Assessment, 23*(1), 166-185. doi:10.1177/1069072714523251	300	College students	22	63.7	Not given	1-5
Miller, J. D., Gentile, B., & Campbell, W. K. (2013). A test of the construct validity of the Five-Factor Narcissism Inventory. *Journal of personality assessment, 95*(4), 377-387. doi:10.1080/00223891.2012.742903	287	Undergraduate students	40	80	18.9	6, 8, 10, 13
Muris, P., Meesters, C., & Timmermans, A. (2013). Some youths have a gloomy side: Correlates of the dark triad personality traits in non-clinical adolescents. *Child Psychiatry & Human Development, 44*(5), 658-665. doi:10.1007/s10578-013-0359-9	117	Dutch adolescents	43	Not given	13.9	1-6
Nagler, U. K., Reiter, K. J., Furtner, M. R., & Rauthmann, J. F. (2014). Is there a "dark intelligence"? Emotional intelligence is used by dark personalities to emotionally manipulate others. *Personality and individual differences, 65*, 47-52. doi:10.1016/j.paid.2014.01.025	594	University students	23	Not given	22.65	4
Noser, A. E., Zeigler-Hill, V., & Besser, A. (2014). Stress and affective experiences: The importance of dark personality features. *Journal of Research in Personality, 53*, 158-164. doi:10.1016/j.jrp.2014.10.007	193	Community adults	49	Israeli (100)	33.84	
Onley, M., Veselka, L., Schermer, J. A., & Vernon, P. A. (2013). Survival of the Scheming: A Genetically Informed Link Between the Dark Triad and Mental Toughness. *Twin Research and Human Genetics, 16*(6), 1087-1095. doi:10.1017/thg.2013.66	420	Adult twins	13	Not given	41.42	

Pabian, S., De Backer, C. J., & Vandebosch, H. (2015). Dark Triad personality traits and adolescent cyber-aggression. *Personality and Individual Differences, 75*, 41-46. doi:10.1016/j.paid.2014.11.015	324	High-school adolescents	37	Not given	16.05	6
Pailing, A., Boon, J., & Egan, V. (2014). Personality, the Dark Triad and violence. *Personality and individual differences, 67*, 81-86. doi:10.1016/j.paid.2013.11.018	159	Community adults	27	Not given	29.25	1-6, 10
Paulhus, D. L., & Williams, K. M. (2002). The dark triad of personality: Narcissism, Machiavellianism, and psychopathy. *Journal of research in personality, 36*(6), 556-563. doi:10.1016/S0092-6566(02)00505-6	245	University students	35	Not given	Not given	1-5
Petrides, K. V., Vernon, P. A., Schermer, J. A., & Veselka, L. (2011). Trait emotional intelligence and the dark triad traits of personality. *Twin Research and Human Genetics, 14*(1), 35-41. doi:10.1375/twin.14.1.35	428	Community adults	14	Not given	41.17	4, 9
Porter, S., Bhanwer, A., Woodworth, M., & Black, P. J. (2014). Soldiers of misfortune: An examination of the Dark Triad and the experience of schadenfreude. *Personality and Individual Differences, 67*, 64-68. doi:10.1016/j.paid.2013.11.014	120	Undergraduate students	27	Not given	20.55	
Rasmussen, K. R., & Boon, S. D. (2014). Romantic revenge and the Dark Triad: A model of impellance and inhibition. *Personality and individual differences, 56*, 51-56. doi:10.1016/j.paid.2013.08.018	165	University students	39	60	20.81	
Rauthmann, J. F. (2011). Acquisitive or protective self-presentation of dark personalities? Associations among the Dark Triad and self-monitoring. *Personality and Individual Differences, 51*(4), 502-508. doi:10.1016/j.paid.2011.05.008	408	University students	22	Not given	22.81	
Rauthmann, J. F. (2012). The Dark Triad and interpersonal perception: Similarities and differences in the social consequences of narcissism, Machiavellianism, and psychopathy. *Social Psychological and Personality Science, 3*(4), 487-496. doi:10.1177/1948550611427608	148	College students	27		22.49	1-5, 13

Rauthmann, J. F., & Kolar, G. P. (2013). Positioning the Dark Triad in the interpersonal circumplex: The friendly-dominant narcissist, hostile-submissive Machiavellian, and hostile-dominant psychopath?. *Personality and individual differences, 54*(5), 622-627. doi:10.1016/j.paid.2012.11.021	184	College students	48	Not given	23.78	
Rauthmann, J. F., & Kolar, G. P. (2013). Positioning the Dark Triad in the interpersonal circumplex: The friendly-dominant narcissist, hostile-submissive Machiavellian, and hostile-dominant psychopath?. *Personality and individual differences, 54*(5), 622-627. doi:10.1016/j.paid.2012.11.021	186	College students	26.8	Not given	22.49	
Saklofske, D. H., Austin, E. J., Yan, G., & Smith, M. M. (2015). Factor structure and correlates of the Mandarin version of the Managing the Emotions of Others (MEOS) scale. *International Journal of Psychology, n/a*, n/a. doi:10.1002/ijop.12147	277	Chinese university students	19	Not given	21.02	1-5, 9
Semenyna, S. W., & Honey, P. L. (2015). Dominance styles mediate sex differences in Dark Triad traits. *Personality and Individual Differences, 83*, 37-43. doi:10.1016/j.paid.2015.03.046	300	Undergraduate students	26	Not given	20.7	13
Semenyna, S. W., & Honey, P. L. (2015). Dominance styles mediate sex differences in Dark Triad traits. *Personality and Individual Differences, 83*, 37-43. doi:10.1016/j.paid.2015.03.046	286	Undergraduate students	33	Not given	20.41	13
Smith, C. V., Hadden, B. W., Webster, G. D., Jonason, P. K., Gesselman, A. N., & Crysel, L. C. (2014). Mutually attracted or repulsed? Actor–partner interdependence models of Dark Triad traits and relationship outcomes. *Personality and Individual Differences, 67*, 35-41. doi:10.1016/j.paid.2014.01.044	45	Undergraduate couple members	0 (all women sample)	79	21.5	
Smith, C. V., Hadden, B. W., Webster, G. D., Jonason, P. K., Gesselman, A. N., & Crysel, L. C. (2014). Mutually attracted or repulsed? Actor–partner interdependence models of Dark Triad traits and relationship outcomes. *Personality and Individual Differences, 67*, 35-41. doi:10.1016/j.paid.2014.01.044	45	Undergraduate couple members	100	79	21.5	
Stead, R., & Fekken, G. C. (2014). Agreeableness at the Core of the Dark Triad of Personality. *Individual Differences Research,*	548	MTurk	29	Not given	32.05	2, 4, 6-7, 9-10

12(4-A), 131–141.						
Stead, R., Fekken, G. C., Kay, A., & McDermott, K. (2012). Conceptualizing the dark triad of personality: Links to social symptomatology. *Personality and Individual Differences, 53*(8), 1023-1028. doi:10.1016/j.paid.2012.07.021	198	University students	23	Not given	20.26	1-5, 14
Stead, R., Fekken, G. C., Kay, A., & McDermott, K. (2012). Conceptualizing the dark triad of personality: Links to social symptomatology. *Personality and Individual Differences, 53*(8), 1023-1028. doi:10.1016/j.paid.2012.07.021	232	University students	13	Not given	19.5	1-5, 14
Stellwagen, K. K., & Kerig, P. K. (2013). Dark triad personality traits and theory of mind among school-age children. *Personality and Individual Differences, 54*(1), 123-127. doi:10.1016/j.paid.2012.08.019	146	Community middle school students	42	82	Middle School Children (6-8th grade)	
Stoeber, J. (2014). How other-oriented perfectionism differs from self-oriented and socially prescribed perfectionism. *Journal of Psychopathology and Behavioral Assessment, 36*(2), 329-338. doi:10.1007/s10862-013-9397-7	338	University students	19	Not given	19.8	
Turnipseed, D. L., & Cohen, S. R. (2015). Academic entitlement and socially aversive personalities: Does the Dark Triad predict academic entitlement?. *Personality and Individual Differences, 82*, 72-75. doi:10.1016/j.paid.2015.03.003	169	Business students	38	Not given	22.5	
Vernon, P. A., Villani, V. C., Vickers, L. C., & Harris, J. A. (2008). A behavioral genetic investigation of the Dark Triad and the Big 5. *Personality and Individual Differences, 44*(2), 445-452. doi:10.1016/j.paid.2007.09.007	278	Community adults	22	Not given	41.42	1-5
Veselka, L., Giammarco, E. A., & Vernon, P. A. (2014). The Dark Triad and the seven deadly sins. *Personality and individual differences, 67*, 75-80. doi:10.1016/j.paid.2014.01.055	253	Undergraduate students	22	Not given	21.99	
Veselka, L., Schermer, J. A., & Vernon, P. A. (2011). Beyond the big five: The dark triad and the supernumerary personality inventory. *Twin Research and Human Genetics, 14*(2), 158-168. doi:10.1375/twin.14.2.150	912	Community adults	30	Not Given	41.42	8

Veselka, L., Schermer, J. A., Martin, R. A., & Vernon, P. A. (2010). Relations between humor styles and the Dark Triad traits of personality. *Personality and Individual Differences, 48*(6), 772-774. doi:10.1016/j.paid.2010.01.017	228	Community adults	15	Not given	41.4	6, 10
Visser, B. A., Pozzebon, J. A., & Reina-Tamayo, A. M. (2014). Status-driven risk taking: Another "dark" personality?. *Canadian Journal of Behavioural Science*, *46*(4), 485. doi:10.1037/a0034163	178	University students	48	Not given	22.13	1-5, 8
Visser, B. A., Pozzebon, J. A., & Reina-Tamayo, A. M. (2014). Status-driven risk taking: Another "dark" personality?. *Canadian Journal of Behavioural Science*, *46*(4), 485. doi:10.1037/a0034163	401	Undergraduate students	22	Not Given	20.47	1-5, 8
Visser, B. A., Pozzebon, J. A., & Reina-Tamayo, A. M. (2014). Status-driven risk taking: Another "dark" personality?. *Canadian Journal of Behavioural Science*, *46*(4), 485. doi:10.1037/a0034163	166	Undergraduate students	29	Not Given	20.15	8, 10
Wai, M., & Tiliopoulos, N. (2012). The affective and cognitive empathic nature of the dark triad of personality. *Personality and Individual Differences, 52*(7), 794-799. doi:10.1016/j.paid.2012.01.008	139	University students	23	Not given	19.9	10
Williams, K. M., Nathanson, C., & Paulhus, D. L. (2010). Identifying and profiling scholastic cheaters: their personality, cognitive ability, and motivation. *Journal of Experimental Psychology: Applied, 16*(3), 293-307. doi:10.1037/a0020773	249	Undergraduate students	30	41.4	Not given	1-5
Williams, K. M., Nathanson, C., & Paulhus, D. L. (2010). Identifying and profiling scholastic cheaters: their personality, cognitive ability, and motivation. *Journal of Experimental Psychology: Applied, 16*(3), 293-307. doi:10.1037/a0020773	107	Undergraduate students	33	43	Not given	1-5
Wilson, M. S., & Hartley, S. M. (2015). Narcissism, Psychopathy and Machiavellianism: Associations between cultural factors and interpersonal dominance. In M. Fitzgerald, M. Fitzgerald (Eds.), *Psychopathy: Risk factors, behavioral symptoms and treatment options* (pp. 75-93). Hauppauge, NY, US: Nova Science Publishers.	185	Undergraduate students	30	73	19.3	

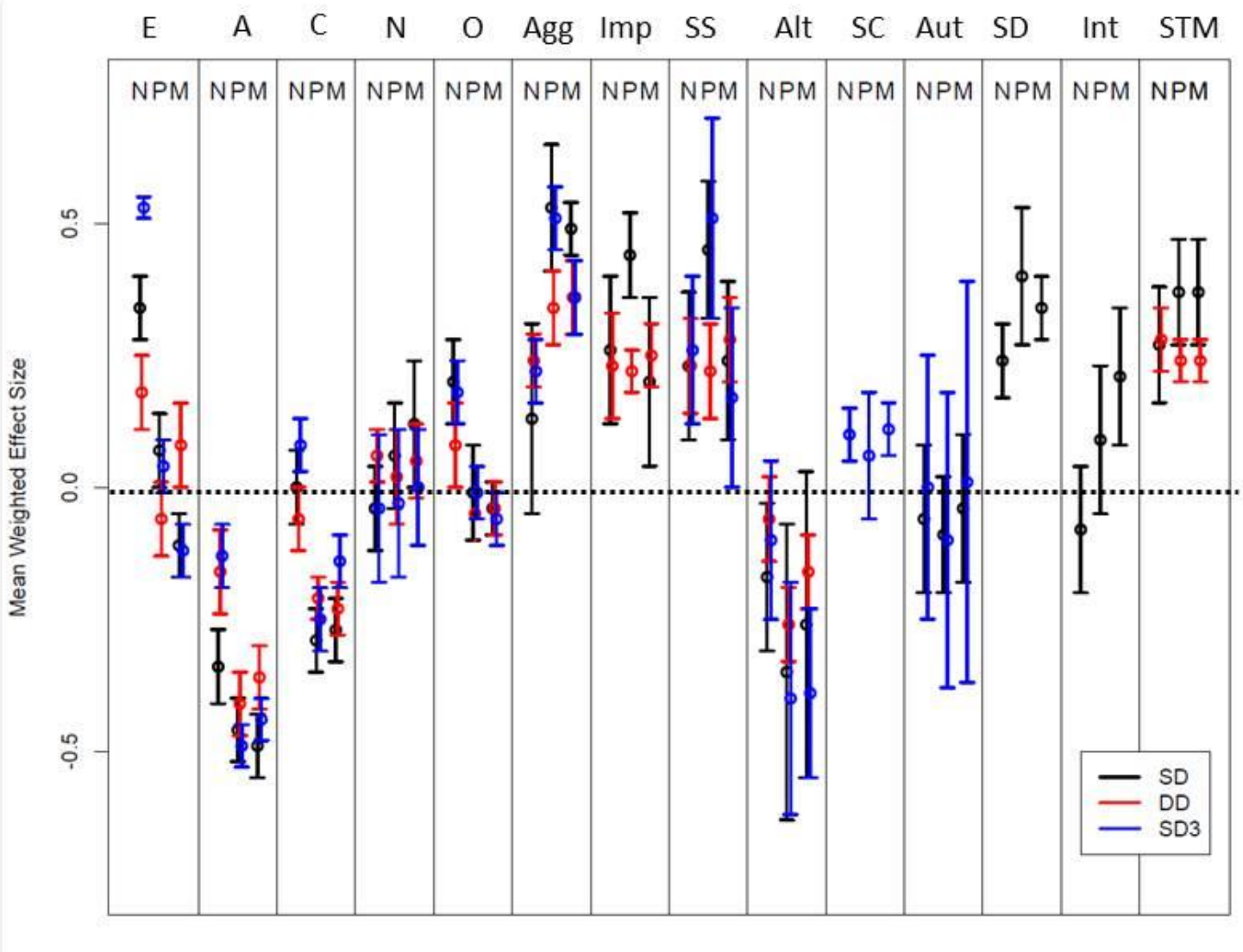
Wisse, B., Barelds, D. P., & Rietzschel, E. F. (2015). How innovative is your employee? The role of employee and supervisor Dark Triad personality traits in supervisor perceptions of employee innovative behavior. *Personality and Individual Differences, 82*, 158-162. doi:10.1016/j.paid.2015.03.020	299	Dutch employees	43.3	Not given	34.44	
Wisse, B., Barelds, D. P., & Rietzschel, E. F. (2015). How innovative is your employee? The role of employee and supervisor Dark Triad personality traits in supervisor perceptions of employee innovative behavior. *Personality and Individual Differences, 82*, 158-162. doi:10.1016/j.paid.2015.03.020	300	Dutch supervisors	58.2	Not given	42.56	
Woodley, H. J., & Allen, N. J. (2014). The dark side of equity sensitivity. *Personality and Individual Differences, 67*, 103-108. doi:10.1016/j.paid.2014.03.003	619	University students	29	57	18	
Zhang, W., Zou, H., Wang, M., & Finy, M. S. (2015). The role of the Dark Triad traits and two constructs of emotional intelligence on loneliness in adolescents. *Personality and Individual Differences, 75*, 74-79. doi:10.1016/j.paid.2014.10.025	396	Chinese adolescents	43	0	15.82	9

Note: The outcomes corresponding to the numbers in the final column are as follows: 1 = Extraversion; 2 = Agreeableness; 3 = Conscientiousness; 4 = Neuroticism; 5 = Openness to Experience; 6 = aggression; 7 = impulsivity; 8 = sensation seeking/risk taking; 9 = emotional intelligence; 10 = altruism/empathy; 11 = social conservatism; 12 = authoritarianism; 13 = social dominance; 14 = internalizing symptoms; 15 = short-term mating behaviors.

Supplemental Figure 1.

Mean Effect Sizes with Confidence Intervals as a Function of Construct and Measure.

Note: E = Extraversion; A = Agreeableness, C = Conscientiousness; N = Neuroticism; O = Openness; Agg = Aggression; Imp = Impulsivity; SS = Sensation Seeking/Risk-Taking; Alt = Altruism; SC = Social Closeness; Aut = Authoritarianism; SD = Social Dominance; Int = Internalizing; STM = Short-term Mating Strategies. Within each column, N = Narcissism, P = Psychopathy, and M = Machiavellianism. Different measures are indexed but different colors; SD = single dimension scales (e.g., the SRP, Mach IV, or NPI); DD = Dirty Dozen; SD3 = Short Dark Triad.



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